

World's Smallest Programmable INDUSPHONE DESIGN

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This is the second part of designing the world's smallest phone, where the UI is integrated with the basic functions of the phone. Previously, the Indusphone user interface (UI) was designed and the GSM module was tested, integrating it with the IndusBoard coin to advance the design. This segment combines the UI with such functions as contacts and calculator. The connections established in part 1, published in EFY's October issue, remain unchanged. The focus now shifts to software and UI integration for the IndusBoard phone.

The initial step involves integrating fundamental functions like dialling a number, making calls, checking call status, handling incoming calls, and displaying the idle screen within the previously designed UI. Fig. 1 illustrates the IndusBoard phone.

Integrating the calling system

In the code, the GSM call library and the TFT_eSPI libraries combine into

a single codebase. An idle screen is created to monitor incoming calls, battery status, and signal strength in a continuous loop. When a call comes in, the UI transitions

to display the 'attend' and 'end' buttons. The function to attend the call is activated when the attend button is touched on the UI. The code checks the UI input; if the input is 'attend' or 'end,' it navigates to the corresponding screen and executes the appropriate function using the 'call.answer();' and 'call.hangup();' commands. Fig. 2 provides a code snippet for TFT_eSPI, while Fig. 3 depicts the initial

```
GSMtest.ino  Free_Fonts.h
#include <TFT_eSPI.h>
#include <CST816S.h>
#include "Free_Fonts.h"
#include <Wire.h>
#include <TimeLib.h> // Add Time library for real-time clock
// =====
#include <GSMSimCall.h>
// Initialize a HardwareSerial instance
HardwareSerial GSMSerial(1);
#define RESET_PIN 10 // you can use any pin.
static volatile int num = 0;
GSMSimCall call(GSMSerial, RESET_PIN); // GSMSimCall inherits
```

Fig. 2: Code snippet for TFT_eSPI

```
void loop() {
  if (onInitialScreen) {
    // Transition to incoming call screen after a delay (e.g., 3 seconds)
    delay(4000); // 3-second delay
    onInitialScreen = false;
    tft.fillScreen(TFT_BLACK); // Clear the screen
    drawPhoneNumber();
    drawButtons(false); // First screen: Both buttons
  }

  if (touch.available()) {
    int x = touch.data.x;
    int y = touch.data.y;

    if (isCallButtonTouched(x, y) && !isCallActive) { // Start call if not active
      Serial.println("Call button pressed");
      startCall();
    }

    if (isEndButtonTouched(x, y) && isCallActive) { // End call only if active
      Serial.println("End button pressed");
      endCall();
    }
  }

  if (isCallActive) {
    updateCallTimer();
  }
}
```

Fig. 3: Initial idle screen code checking for incoming calls, network, and battery levels



Fig. 1: IndusBoard phone